Fairness, and Ethics

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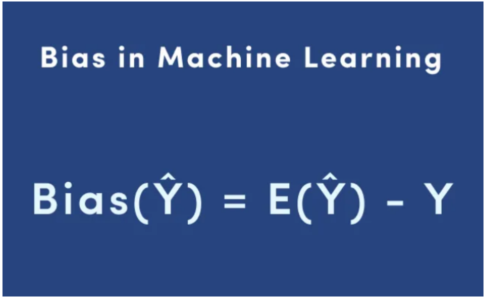
November 10, 2025

Introduction to Bias in ML

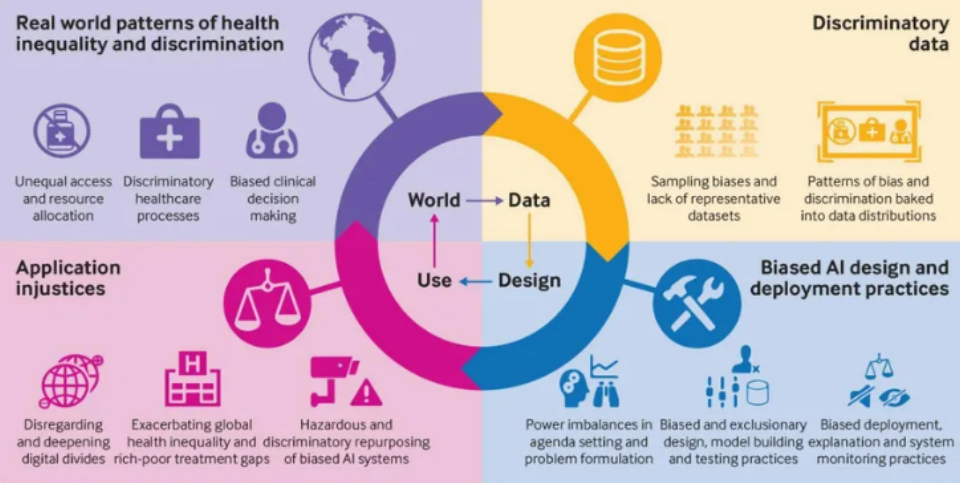
- Bias = systematic error that disadvantages groups
- Can come from data, algorithms, or human choices
- Example: Skin tone bias in dermatology apps (link).
- A 2023 study introduced the STAR-ED framework to analyze thousands of dermatology images in textbooks, lecture slides, and journal articles. It found that only about 10% of images had dark skin tones (Fitzpatrick types V–VI), highlighting serious representation bias.

Types of Bias

- Sample bias
- Label bias
- Measurement bias
- Historical bias



Impact of Bias



Case Study – Algorithmic Bias

- Correction factor in kidney function algorithms disadvantaging Black patients (link)
  - Clinical formulas used to estimate kidney function (eGFR)—such as the widely used CKD-EPI and MDRD equations—included a multiplicative correction for Black race. This adjustment increased estimated kidney function by up to 16–21% for Black individuals with the same creatinine, age, and sex
  - Major medical centers have since eliminated the race adjustment, adopting race-neutral eGFR formulas.
- Showed how embedded bias can persist and affect treatment.

Case Study – Algorithmic Bias

- Amazon's Biased Recruiting Algorithm (2014–2017) (link)
  - **What happened:** An internal AI tool used by Amazon to screen résumés was found to downgrade female applicants due to historical bias in hiring data.
  - **Ethical failures:** Biased training data led to discrimination, lack of auditing before deployment.
  - **Legacy:** A cautionary tale about how biased inputs yield biased outputs in ML.

Detecting and Measuring Bias

- **Fairness metrics**
  - **Demographic parity**:

    $$P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$$

    Ensures equal positive prediction rates across groups (e.g., race, gender).
  - **Equal opportunity / Equalized odds**:

    $$P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$$

    Measures equality of sensitivity (or both TPR/FPR) across protected groups.
  - Other metrics: predictive parity (equal PPV), calibration within groups.

Detecting and Measuring Bias

- **Fairness metrics**
  - **Comparing model performance across groups**
    - Compute AUC, accuracy, sensitivity, and specificity stratified by protected attribute (e.g., sex, race, income level).
    - Use bootstrap confidence intervals to assess significant differences.
    - Evaluate calibration curves and residuals separately for each subgroup.
- **Auditing datasets and models**
  - Examine feature distributions and label frequencies by group.
    (e.g., Are certain subpopulations underrepresented or mislabeled?)
  - Use tools such as `Aequitas`, `Fairlearn`, or `Themis-ML` to compute fairness diagnostics automatically.
  - Conduct statistical tests (e.g., chi-squared or permutation tests) for group-level disparities in predictions.

Mitigating Bias

- **Diverse, representative data collection**
  - Oversample underrepresented groups (e.g., SMOTE, stratified sampling).
  - Balance cohorts by key demographics during data acquisition.
  - Use reweighting schemes to equalize sample contributions: assign weights $w_i \propto 1/P(A_i)$.

Mitigating Bias

- **Fairness-aware ML algorithms**
  - **Pre-processing methods:**
    - Reweighing (Kamiran & Calders, 2012): adjust sample weights to remove group–label dependence.
    - Disparate impact remover: transform features to reduce correlation with protected attributes.
  - **In-processing methods:**
    - Add fairness constraints or penalties to the loss function, e.g.: $\text{Loss} + \lambda|\text{TPR}_{A=0} - \text{TPR}_{A=1}|$
    - Adversarial debiasing: train an auxiliary network to predict the protected attribute and penalize its accuracy.

Mitigating Bias

- **Fairness-aware ML algorithms**
  - **Post-processing methods:**
    - Calibrate decision thresholds separately for each group to equalize TPR or FPR (Hardt et al., 2016).
    - Reject option classification: modify predictions for uncertain cases to favor fairness.
- **Post-deployment monitoring**
  - Continuously audit model performance as population characteristics shift.
  - Evaluate fairness metrics on new data to detect drift or emerging inequities.

Choosing an Approach

Situation	Preferred Method	Key Benefit
Rare outcome overall	SMOTE	Improves minority-class detection
Underrepresented groups	Reweighting	Equalizes influence in training
Unequal performance after training	Group thresholds	Equalizes TPR/FPR across groups

- In public health models, fairness means ensuring comparable utility for all subpopulations, not necessarily identical predictions.

What is Data Privacy?

- Protecting individuals' personal and health data from misuse
- Key concept: Identifiability
- Example: De-identified vs anonymized data

Legal and Ethical Standards

- HIPAA (U.S.) and GDPR (Europe)
- Institutional Review Boards (IRBs).
- Duty to protect participant confidentiality

Security in Machine Learning

- Storing and transmitting data securely
- Encryption, access control, and audit trails
- Risk of model inversion and membership inference attacks

Case Study – Privacy Breach

- In 1997, researcher Latanya Sweeney successfully re-identified Governor William Weld's "anonymized" medical records by linking a de-identified hospital dataset with publicly accessible voter registration information.
- She used just three quasi-identifiers—ZIP code, birth date, and gender—to uniquely match his hospital visits in Cambridge, Massachusetts.

Informed Consent in ML

- Not just checkboxes: meaningful understanding
- Include ML use in study protocols and consent forms
- Publicly available model documentation

Case Study – Lack of Consent

- Henrietta Lacks and HeLa Cells (1951)
  - **What happened:** Cells from Henrietta Lacks, a Black woman with cervical cancer, were taken without her knowledge or consent. They became the first immortal human cell line and were used in countless scientific breakthroughs.
  - **Ethical failures:** No consent, no compensation to the family, lack of privacy.
  - **Legacy:** Raised awareness about ownership of biological data and informed consent in biomedical research.

Case Study – Privacy Breach

- Re-identification of Netflix user data (link)
  - Researchers Arvind Narayanan and Vitaly Shmatikov demonstrated that it was possible to de-anonymize users in the Netflix Prize dataset by matching a small number of their movie ratings and dates with records from IMDb. Even minimal auxiliary information (e.g., 8 ratings with approximate dates) enabled 99% re-identification of users, exposing sensitive personal details
- ML models can unintentionally leak private info
- Importance of privacy-preserving techniques

Case Study – Lack of Consent

- Cambridge Analytica and Facebook data scandal (link)
- Consent was not truly informed
- Highlights the importance of transparency and user control

Facial Recognition in Public Health Monitoring

- Facial recognition technology has been used for:
  - Contactless temperature checks during COVID-19
  - Identifying individuals violating quarantine orders
  - Monitoring mask compliance in public spaces
- Raises serious concerns about privacy, surveillance, and consent.
- **Case Example:** Use in China for monitoring quarantine enforcement.

NYT: China Turns to Surveillance to Fight Coronavirus

Strava Heatmap Reveals Military Bases

- In 2018, fitness app Strava released a global heatmap of user activity.
- Publicly available map unintentionally revealed the locations of:
  - Secret military bases and patrol routes
  - Sensitive national security sites in conflict zones
- Raises questions about:
  - How "anonymous" data can be de-anonymized
  - The need for better data governance in health apps

BBC: Fitness app Strava lights up staff at military bases

Ethics in the ML Lifecycle

- Ethical considerations at every stage:
  - Data collection
  - Model training
  - Deployment
  - Monitoring and updating

Your Role as Future Public Health Leaders

- Ask tough questions about fairness and privacy.
- Involve communities and ethicists from the start
- Build fairness checks into pipelines
- Provide clear explanations of models and decision processes
- Use ML to help, not harm.

Resources and Further Reading

- https://www.fatml.org/
- *Weapons of Math Destruction* by Cathy O'Neil
- https://www.partnershiponai.org/
- WHO guidance on AI ethics in health